DISTINCT IRREDUCIBLE CLOSED SUBSETS OF AN INTEGRAL SEPARATED SCHEME HAVE DISTINCT LOCAL RINGS IN THE FUNCTION FIELD

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Local rings of an integral scheme are local subrings of the function field. Lemma \ref{lemma-integral-scheme-rational-functions}, cited at proof lines 12752-12754. Makes the two local rings comparable as subrings of one field.
- A unique morphism to the scheme from the spectrum of the local ring, compatible with the canonical map from the function field. Schemes, Equation \ref{schemes-equation-canonical-morphism} and Schemes, Lemma \ref{schemes-lemma-where-are-they-equal}, cited at proof lines 12773-12785. Separatedness forces that morphism to be unique.
- Specialisation of points read off from a morphism of spectra. Schemes, Lemma \ref{schemes-lemma-specialize-points}, cited at proof lines 12799-12801. Turns the hypothetical inclusion into a containment of the closures.

Source reference. Original lemma 02NF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3478–3478. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 02NA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3473–3473. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01KM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2055–2055. Navigation pointer to a statement the

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Source reference. Original lemma 01J7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2004–2004. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let X be an integral separated scheme. Let Z_1, Z_2 be distinct irreducible closed subsets of X. Let η_i be the generic point of Z_i. If $Z_1 \not\subset Z_2$, then $\mathcal{O}_{X,\eta_1} \not\subset \mathcal{O}_{X,\eta_2}$ as subrings of $R(X)$. In particular, if $Z_1 = \{x\}$ consists of one closed point x, there exists a function regular in a neighborhood of x which is not in $\mathcal{O}_{X,\eta_2}$.*

Proof. First observe that under the assumption of X being separated, there is a unique map of schemes $\mathrm{Spec}(\mathcal{O}_{X,\eta_2}) \to X$ over X such that the composition

$$\mathrm{Spec}(R(X)) \longrightarrow \mathrm{Spec}(\mathcal{O}_{X,\eta_2}) \longrightarrow X$$

is the canonical map $\mathrm{Spec}(R(X)) \to X$. Namely, there is the canonical map $can : \mathrm{Spec}(\mathcal{O}_{X,\eta_2}) \to X$, see Schemes, Equation (02NA). Given a second morphism a to X, we have that a agrees with can on the generic point of $\mathrm{Spec}(\mathcal{O}_{X,\eta_2})$ by assumption. Now X being separated guarantees that the subset in $\mathrm{Spec}(\mathcal{O}_{X,\eta_2})$ where these two maps agree is closed, see Schemes, Lemma 01KM. Hence $a = can$ on all of $\mathrm{Spec}(\mathcal{O}_{X,\eta_2})$.

Assume $Z_1 \not\subset Z_2$ and assume on the contrary that $\mathcal{O}_{X,\eta_1} \subset \mathcal{O}_{X,\eta_2}$ as subrings of $R(X)$. Then we would obtain a second morphism

$$\mathrm{Spec}(\mathcal{O}_{X,\eta_2}) \longrightarrow \mathrm{Spec}(\mathcal{O}_{X,\eta_1}) \longrightarrow X.$$

By the above this composition would have to be equal to can. This implies that η_2 specializes to η_1 (see Schemes, Lemma 01J7). But this contradicts our assumption $Z_1 \not\subset Z_2$. □